

The study area contains a variety of land-cover types, including urban areas, vegetation, bare soil, and water (Fig. 1). The training samples are spatially distributed across the study area rather than concentrated in a single location (Fig. 2). This distribution was intended to capture the spectral variability within each land-cover class and provide representative training data for the supervised classifiers.

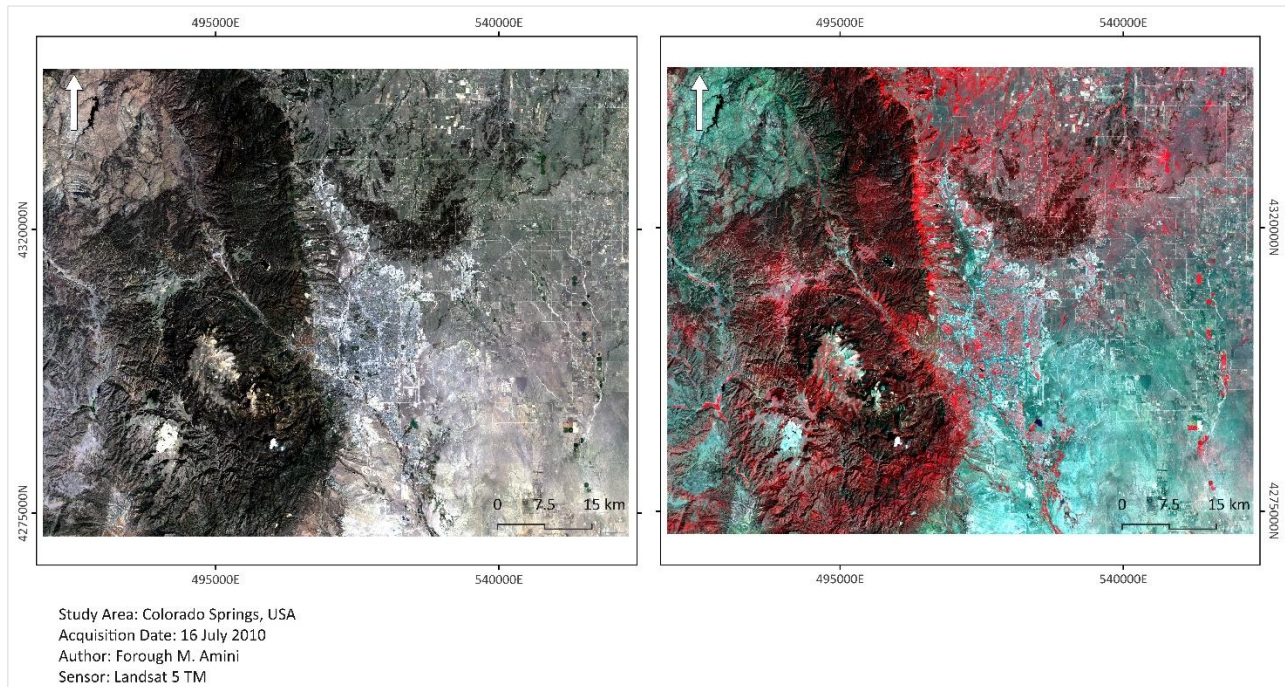


Figure 1: Region of interest, Colorado Springs, USA. True image (left) and False image (right).

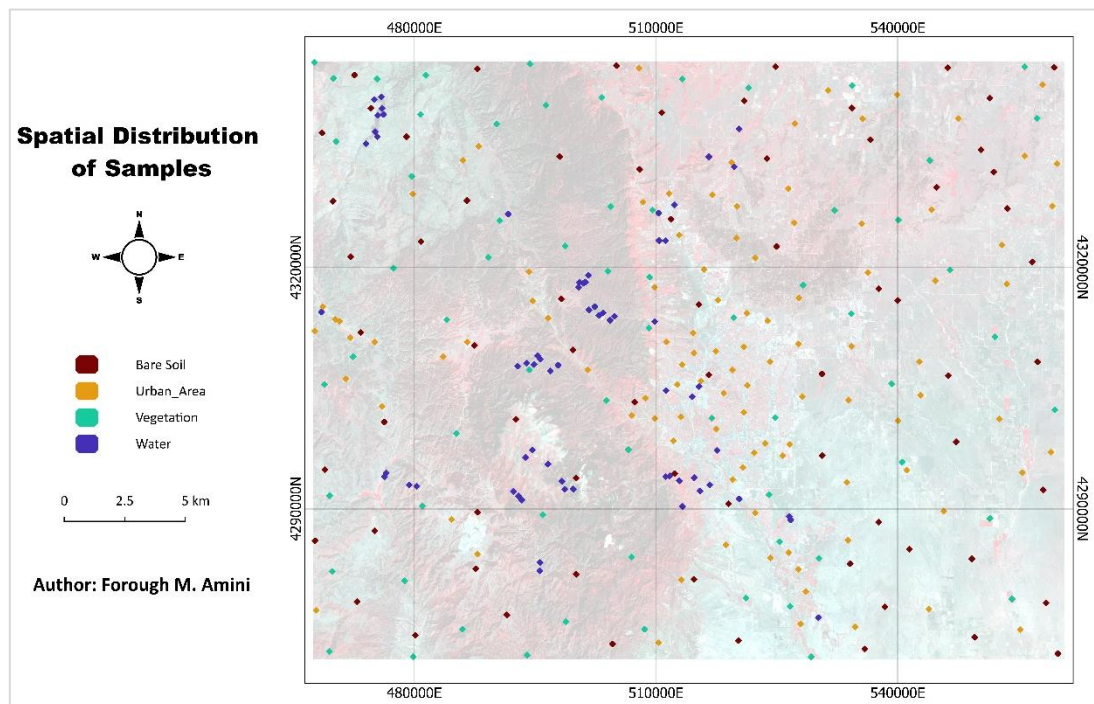


Figure 2: Spatial distribution of samples for all classes.

The baseline classification maps show noticeable differences between the classifiers in the urban region (Fig. 3). Urban areas are generally identified by all models, but confusion with bare soil is visible, particularly in areas where urban surfaces and exposed soil have similar spectral characteristics. The Nearest Centroid, Maximum Likelihood Classifier, and K-Nearest Neighbors produce relatively similar spatial patterns, while the Decision Tree and Random Forest results contain more local variations. The MLP also captures several narrow urban structures and transitions within the region.

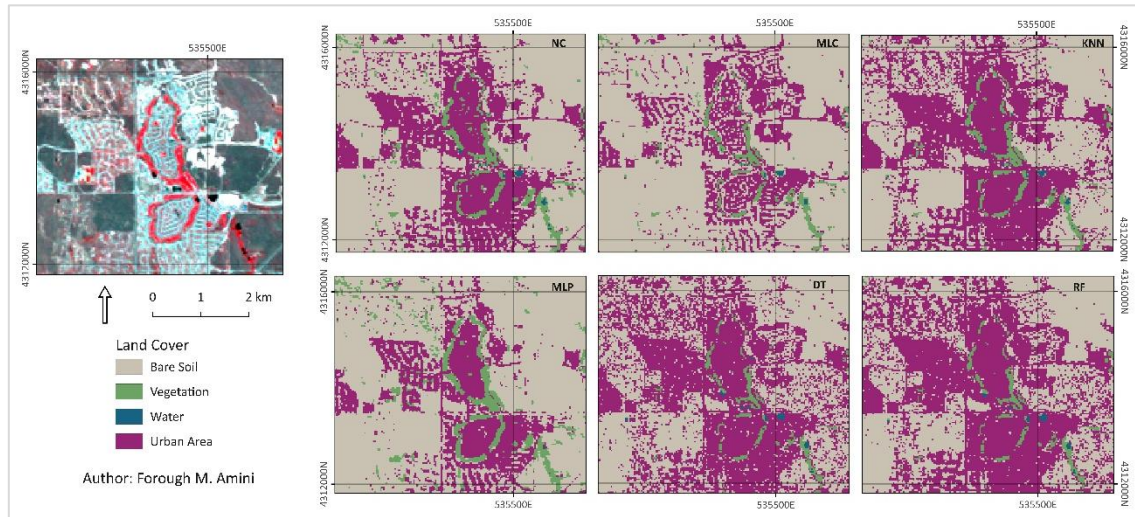


Figure 3: Baseline models' classification results, zooming in on the urban region. NC stands for Nearest Centroid, MLC for Maximum Likelihood Classifier, KNN for K-Nearest Neighbors, MLP for Multi-Layer Perceptron, DT for Decision Tree, and RF for Random Forest.

After hyperparameter tuning, the spatial patterns change differently for each classifier. Some models produce more spatially consistent regions, whereas others show additional local variations. This indicates that hyperparameter optimization does not necessarily lead to a visually cleaner classification for every model. The urban–bare soil confusion remains an important source of classification uncertainty.

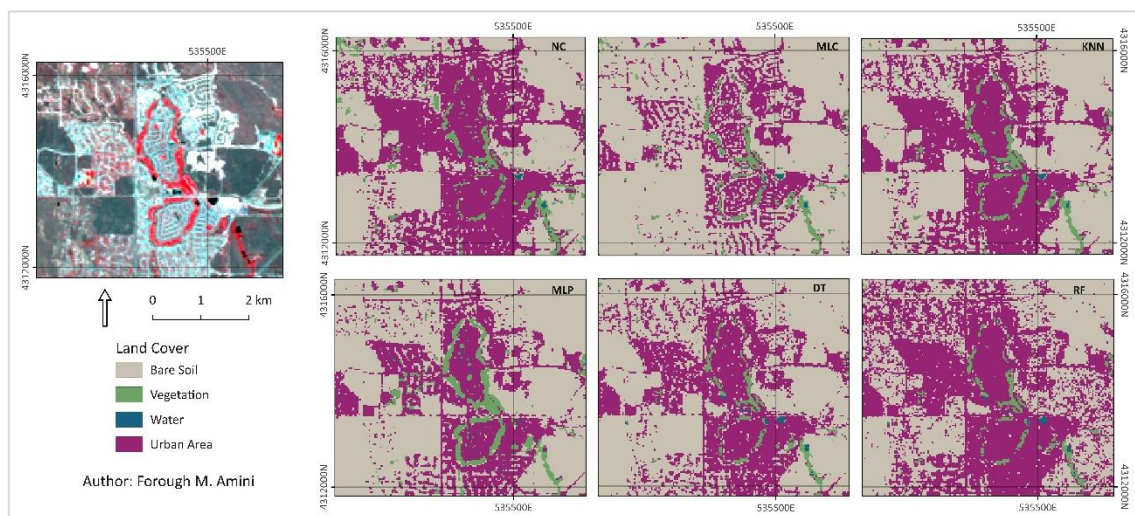


Figure 4: Tuned models' classification results, zooming in on the urban region. NC stands for Nearest Centroid, MLC for Maximum Likelihood Classifier, KNN for K-Nearest Neighbors, MLP for Multi-Layer Perceptron, DT for Decision Tree, and RF for Random Forest.

In the vegetation and bare-soil region, the different classifiers consistently identify the major vegetation structures. However, differences can be observed along the boundaries and in smaller patches. The Decision Tree and Random Forest produce more fragmented patterns in some areas, whereas the simpler classifiers tend to produce more spatially homogeneous regions. These differences reflect the different decision mechanisms used by the classifiers.

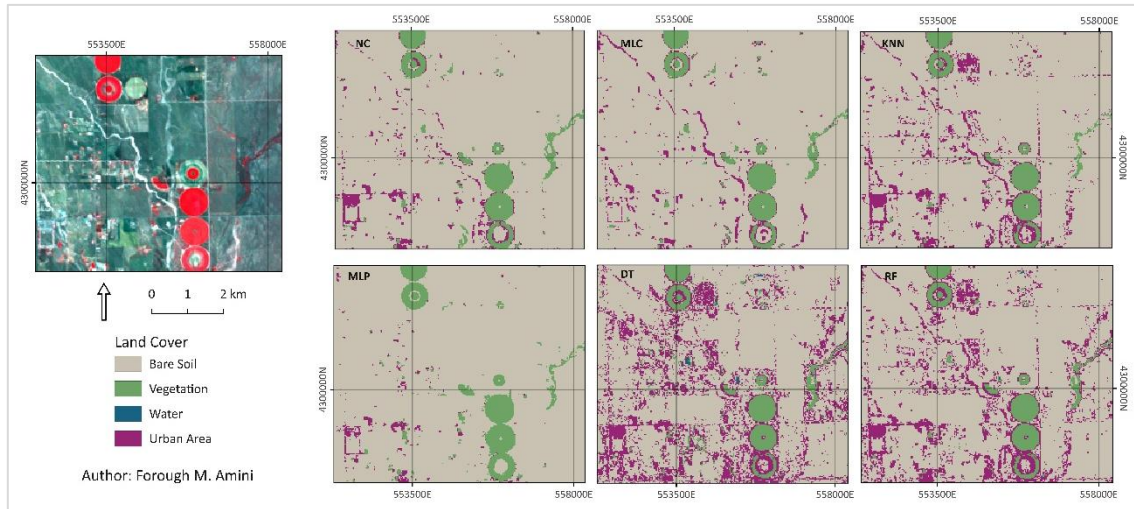


Figure 5: Baseline models' classification results, zooming in on the vegetation and soil regions. NC stands for Nearest Centroid, MLC for Maximum Likelihood Classifier, KNN for K-Nearest Neighbors, MLP for Multi-Layer Perceptron, DT for Decision Tree, and RF for Random Forest.

The tuned models show that changing the model parameters can modify the spatial distribution of the predicted classes. Although some classifications become less fragmented, improvement is not consistent across all models. In particular, the tuned MLP shows substantial changes in the predicted vegetation and water-related areas, illustrating that parameter tuning can also produce less stable spatial results when the available training data are limited.

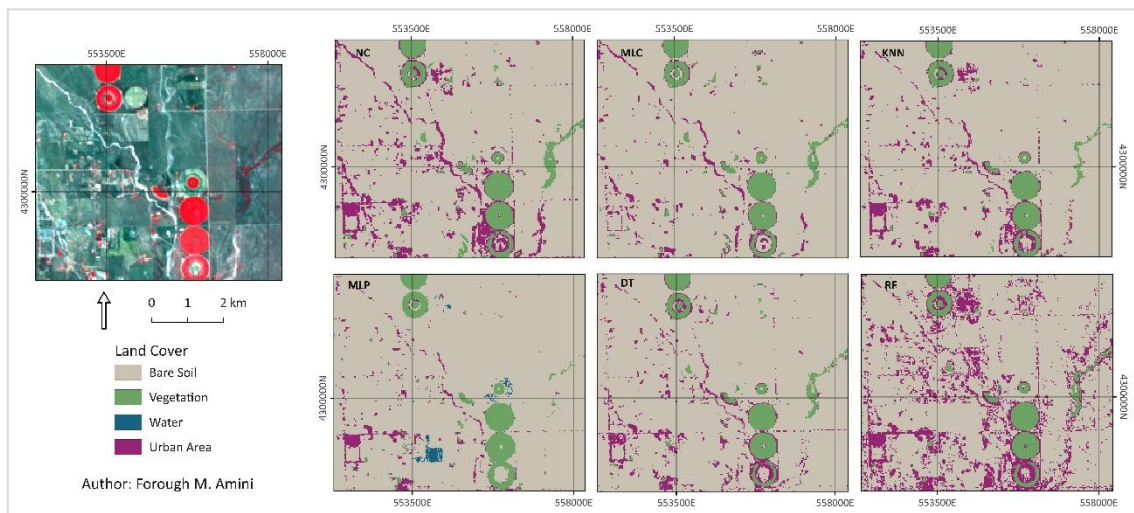


Figure 6: Tuned models' classification results, zooming in on the vegetation and soil regions. NC stands for Nearest Centroid, MLC for Maximum Likelihood Classifier, KNN for K-Nearest Neighbors, MLP for Multi-Layer Perceptron, DT for Decision Tree, and RF for Random Forest.

The water body is identified relatively consistently across the baseline classifications. Its spectral characteristics appear to be more distinct from the surrounding land-cover classes than those of urban and bare-soil areas. Differences between classifiers are therefore more apparent around the boundaries of the water body and in small surrounding areas than in the main water region.

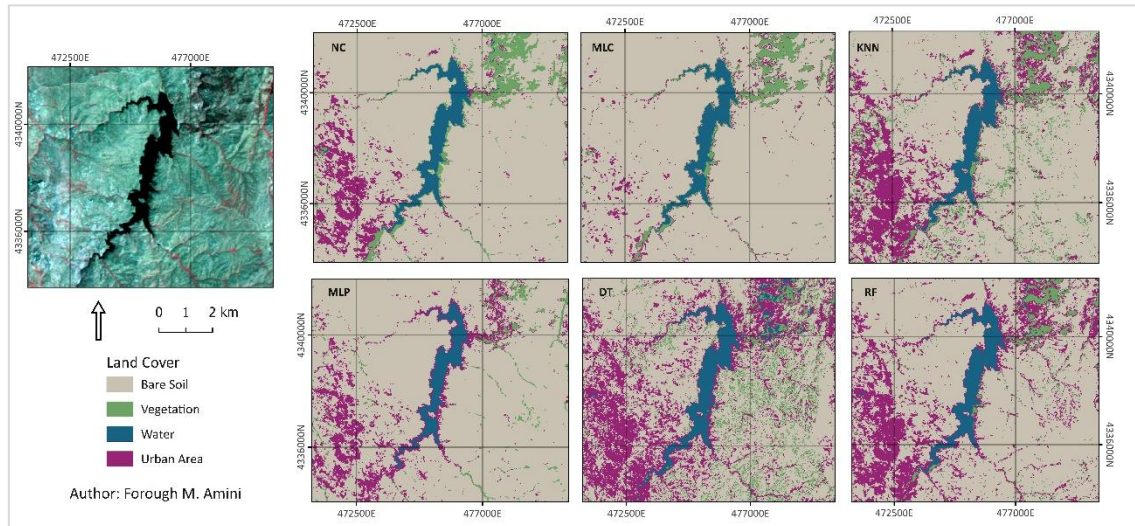


Figure 7: Baseline models' classification results, zooming in on the water resource region. NC stands for Nearest Centroid, MLC for Maximum Likelihood Classifier, KNN for K-Nearest Neighbors, MLP for Multi-Layer Perceptron, DT for Decision Tree, and RF for Random Forest.

The tuned classifications preserve the main water body for most models, but differences remain in the surrounding pixels and along the boundaries. The MLP shows a particularly large change compared with its baseline classification, with additional pixels being assigned to water in areas that are not part of the main water body. This demonstrates that a higher model complexity or tuned parameter configuration does not automatically result in a more spatially coherent classification.

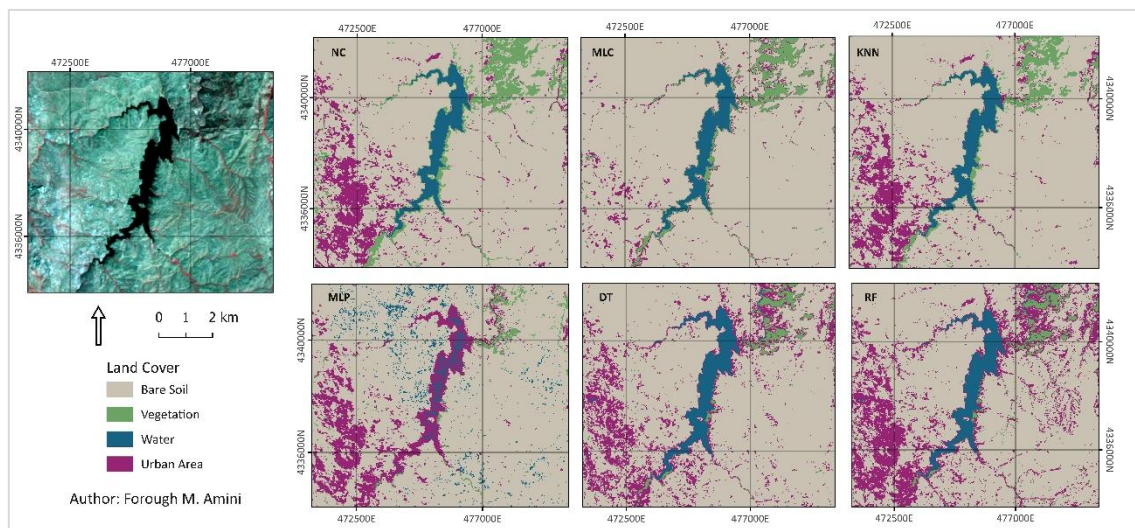


Figure 8: Tuned models' classification results, zooming in on the water resource region. NC stands for Nearest Centroid, MLC for Maximum Likelihood Classifier, KNN for K-Nearest Neighbors, MLP for Multi-Layer Perceptron, DT for Decision Tree, and RF for Random Forest.

Table 1 shows the hyperparameters selected through 10-fold cross-validation. KNN obtained the highest cross-validation accuracy (0.909), followed by Random Forest (0.8851), Nearest Centroid (0.8678), Decision Tree (0.8616), MLP (0.8331), and MLC (0.8318). The selected parameters differ considerably between classifiers, reflecting the different model structures and learning mechanisms.

Table 1. Selected models' hyperparameters through hyperparameter tuning using grid search.

Model	Best Parameters	10-fold cross-validation Accuracy
KNN	n_neighbors=21, p=2, weights=uniform	0.909
RF	bootstrap=False, max_depth=None, max_features=sqrt, min_samples_leaf=2, n_estimators=100	0.8851
NC	metric=manhattan, shrink_threshold=None	0.8678
DT	criterion=entropy, max_depth=7, min_samples_leaf=10, min_samples_split=2	0.8616
MLP	alpha=0.001, hidden_layer_sizes=(12,), learning_rate_init=0.01	0.8331
MLC	var_smoothing=1e-12	0.8318

Table 2 compares the baseline and tuned models on the validation dataset. The effect of hyperparameter tuning was not consistent across the classifiers. KNN and Decision Tree showed an increase in validation accuracy, while Random Forest and Nearest Centroid showed a decrease. MLC remained unchanged because its selected parameter configuration produced the same result as the baseline model. The largest change occurred for MLP, whose validation accuracy decreased substantially after tuning. Overall, these results show that a higher level of model optimization or complexity does not necessarily lead to improved validation performance for this dataset.

Table 2. Accuracy and kappa results estimated for the models and their tuned versions.

Model	Baseline Accuracy	Tuned Accuracy	Δ Accuracy	Baseline Kappa	Tuned Kappa	Δ Kappa
KNN	0.8494	0.8543	0.0049	0.799	0.8055	0.0065
RF	0.8444	0.8309	-0.0136	0.7925	0.7744	-0.0181
NC	0.8346	0.8247	-0.0099	0.7792	0.7661	-0.0131
DT	0.8185	0.8321	0.0136	0.7579	0.7759	0.018
MLP	0.8074	0.6111	-0.1963	0.7428	0.4799	-0.2629
MLC	0.7975	0.7975	0	0.7296	0.7296	0

The producer's and user's accuracies in Tables 3 and 4 provide additional insight into the class-specific behaviour of the models. Water is consistently classified with very high producer's and user's accuracies for most models, indicating that it is spectrally well separated from the other classes in this dataset. Vegetation also shows relatively high accuracies across the classifiers.

In contrast, urban areas show substantially lower producer's accuracies for several models. For the baseline models, urban producer's accuracy ranges from approximately 39% for MLC to 72% for Random Forest, indicating considerable omission of urban pixels. The corresponding user's accuracies are generally higher, but still vary considerably between models. This suggests that urban areas are more difficult to distinguish consistently from other classes, particularly bare soil.

Table 3. Confusion matrix and its accuracy and errors estimated for baseline models. From left to right, the orange colour stands for bare soil, green for vegetation, blue for water, and red for urban classes.

Model	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy
KNN	87.14%	12.86%	71.21%	28.79%	92.42%	7.58%	95.31%	4.69%	98.99%	1.01%	100.00%	0.00%	61.76%	38.24%	76.36%	23.64%
Random Forest	79.52%	20.48%	76.96%	23.04%	87.88%	12.12%	95.60%	4.40%	99.49%	0.51%	100.00%	0.00%	71.57%	28.43%	68.22%	31.78%
Nearest Centroid	94.29%	5.71%	69.72%	30.28%	95.45%	4.55%	86.70%	13.30%	98.99%	1.01%	98.00%	2.00%	45.59%	54.41%	86.11%	13.89%
Decision Tree	73.81%	26.19%	73.11%	26.89%	86.36%	13.64%	95.00%	5.00%	99.49%	0.51%	98.50%	1.50%	68.63%	31.37%	64.22%	35.78%
MLP	95.71%	4.29%	67.45%	32.55%	96.97%	3.03%	88.89%	11.11%	84.85%	15.15%	99.41%	0.59%	45.59%	54.41%	73.23%	26.77%
MLC	90.48%	9.52%	60.90%	39.10%	92.42%	7.58%	94.33%	5.67%	97.98%	2.02%	100.00%	0.00%	38.73%	61.27%	71.82%	28.18%

Bare soil also shows considerable variation between classifiers. For example, the baseline producer's accuracy ranges from approximately 74% for the Decision Tree to 96% for the MLP, while user's accuracy ranges from approximately 61% for MLC to 77% for Random Forest. This variation is consistent with the visual differences observed in the classification maps, particularly in areas where urban surfaces and bare soil occur close together.

Table 4. Confusion matrix and its accuracy and errors estimated for tuned models. From left to right, the orange colour stands for bare soil, green for vegetation, blue for water, and red for urban classes.

Model	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy	Producer Accuracy	Omission Error	User Accuracy	Comission Accuracy
KNN	95.71%	4.29%	69.55%	30.45%	92.42%	7.58%	95.31%	4.69%	98.99%	1.01%	98.99%	1.01%	54.90%	45.10%	85.50%	14.50%
Decision Tree	89.05%	10.95%	69.78%	30.22%	86.87%	13.13%	96.09%	3.91%	99.49%	0.51%	97.52%	2.48%	57.84%	42.16%	73.29%	26.71%
Random Forest	76.19%	23.81%	74.77%	25.23%	87.88%	12.12%	95.60%	4.40%	99.49%	0.51%	100.00%	0.00%	69.61%	30.39%	65.44%	34.56%
Nearest Centroid	85.24%	14.76%	69.92%	30.08%	92.93%	7.07%	86.38%	13.62%	98.99%	1.01%	100.00%	0.00%	53.43%	46.57%	75.17%	24.83%
MLC	90.48%	9.52%	60.90%	39.10%	92.42%	7.58%	94.33%	5.67%	97.98%	2.02%	100.00%	0.00%	38.73%	61.27%	71.82%	28.18%
MLP	89.05%	10.95%	66.31%	33.69%	94.95%	5.05%	92.61%	7.39%	10.10%	89.90%	64.52%	35.48%	49.02%	50.98%	34.01%	65.99%

Overall, the results demonstrate that classifier selection and hyperparameter tuning both influence the quantitative and spatial characteristics of the land-cover classification. Water is the most consistently classified class, with very high producer's and user's accuracies across most models, whereas urban areas show greater confusion and omission. The quantitative results also show that hyperparameter tuning does not guarantee an improvement in validation performance: KNN and Decision Tree improved, while Random Forest and Nearest Centroid decreased, MLC remained unchanged, and MLP showed a substantial decrease. The classification maps similarly demonstrate model-specific spatial differences, particularly around urban–bare soil boundaries and smaller vegetation and water patches. Therefore, visual map interpretation and quantitative accuracy assessment provide complementary information for evaluating the classification results.